



1 Automated Test Suite Summary

Klyvix underwent comprehensive unit testing, integration testing with FastAPI TestClient, end-to-end browser automation with Playwright, and load/stress testing against upstream LLM APIs.

Test Category	Scope & Framework	Count	Result
Backend API Unit Tests	FastAPI endpoints, JWT auth, schema parsing (Pytest)	28 tests	PASSED (100%)
Persona Validation Tests	Prompt compliance, JSON schema outputs, refusals	18 tests	PASSED (100%)
Vector RAG Search Tests	PDF parsing, chunking, Qdrant cosine similarity	12 tests	PASSED (100%)
Multi-LLM Failover Tests	Simulated HTTP 429 rate limit & Groq auto-switch	8 tests	PASSED (100%)
Frontend E2E Tests	Playwright browser automation (Desktop & Mobile)	16 tests	PASSED (100%)

2 Persona-Specific Behavioral Verification

 Study Mentor Validation

Pedagogical Guardrails

Verified Socratic questioning loop: breaks complex concepts into 2-3 pieces and refuses to give full exam answers in direct interactive mode. LaTeX math validated.

 Code Reviewer Validation

Pydantic JSON

100% of responses strictly parsed as valid JSON schemas categorized into Bugs, Security, Performance, and Style with line numbers and severity tags.

 Code Colleague Validation

Pair Programming

Verified non-destructive refactoring: returns complete production code with inline explanations without overwriting functioning logic.

 Document Analyzer Validation

Zero Hallucination

Tested against out-of-document questions: returns "Information not found in provided document" with 0% false positives.

3 Performance & Latency Benchmarks

Metric Measured	Target Benchmark	Observed Production SLA	Status
Time to First Token (TTFT)	< 2.5 seconds	1.18 seconds (Gemini 2.0 Flash)	OPTIMAL
Failover Hot-Swap Time	< 1.0 second	285 milliseconds (Groq Llama 3.3)	OPTIMAL
Vector Similarity Search	< 50 milliseconds	12.4 milliseconds (Qdrant Cloud)	OPTIMAL
Frontend Initial Paint	< 100 milliseconds	38 milliseconds (Vanilla JS)	OPTIMAL

4 Multi-LLM Failover Stress Test Scenarios

Scenario A: Gemini 429 Quota Exhaustion

Simulated continuous burst traffic to trigger rate limit. Backend caught ResourceExhausted exception and seamlessly resumed streaming via Groq Llama 3.3 70B without client disconnect.

Scenario B: Network Latency / Timeout

Injected a 5-second socket hang on primary provider. Timeout interceptor triggered fallback route within 350ms, maintaining continuous user experience.

5 Security, Auth & Vulnerability Audit

Security Dimension	Implementation Standard	Verification Result
API Secret Protection	Google Cloud Run Environment Variables	No hardcoded secrets or keys in codebase / git repository.
Password & Token Security	Passlib bcrypt + HS256 JWT tokens	Stateless, tamper-proof session tokens with expiration enforcement.
CORS & Origin Locking	FastAPI CORSMiddleware	Restricted to deployed Cloud Run origin and local development hosts.
Input Sanitization	DOMPurify + Pydantic Type Guards	Eliminates XSS injection risks in rendered markdown and code snippets.